

MODEL QUESTION PAPER

REVISION-2010
TED (10)-4096

FIFTH SEMESTER DIPLOMA EXAMINATION IN POLYMER TECHNOLOGY

TYRE TECHNOLOGY

(Maximum marks-100)

Time: 3Hours

PART-A

(Maximum marks-10)

I. Answer the following questions in one or two sentences.

(Each question carries 2 marks)

1. Define tyre profile?
2. What do the tyre markings given here signify? 155/70R 13 85 S.
3. Which tread pattern is most suitable for rear wheel of trucks? Why?
4. What are the components of a cycle tyre?
5. Name any four non destructive tests on tyres. (2X5=10)

PART-B

(Maximum marks-30)

II. Answer **any five** of the following questions.

(Each question carries 6 Marks)

1. a). Mention the advantages, and disadvantages of nylon as a tyre cord material. (3)
b). Mention the functions of bead and chafer in tyres. (3)
2. a). What is the basic difference between a radial and bias tyre? Illustrate with a line sketch. (3)
b) What are the essential characteristics of tyre cord materials? (3)
3. a) Design a compound for passenger tyre tread? (3)
b) What are the treatments involved in green tyre preparation? (3)
4. a) How is RFL prepared? (3)
b) Mention the different parts of a bias tyre building machine (3)
5. a) List the advantages and disadvantages of solid tyres. (3)
b) Describe the mandrel curing of cycle tubes. (3)
6. a) List the various materials used in conventional retreading process. (3)

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- b) Design a compound for cushion gum. (3)
7. a) Define the terms tex denier and tenacity. (3)
- b) What is cut tyre analysis? (3)

PART-C

(Maximum marks-60)

Answer **one full** question from each unit

(Each question carries 15 marks)

MODULE-I

- IV. a) Summarise the history of tyres with respect to design and raw materials used. (8)
- b) With a neat sketch explain the design and components of a radial tyre. (7)

OR

- V. a) What are the functions of a pneumatic tyre? Explain. (8)
- b) Mention the various tyre components and explain the functions of each of them. (7)

MODULE-II

- VI. a) Explain the steps involved in tyre cord processing with a flow chart' (8)
- b) Design a suitable formulation for curing bladder and describe its manufacture. (7)

OR

- VII. a) With a flow chart explain the process of manufacturing tyre inner tubes. (8)
- b) Differentiate between autoform presses and Bag-O-Matic presses. (7)

MODULE-III

- VIII. a) Explain the mono band and collapsible drum method for the production of cycle tyre. (10)
- b) Compare the solid tyre with pneumatic tyre. (5)

OR

- IX. a) Explain the cold process retreading. (8)
- b) Design a typical tread compound for conventional retreading. (7)

MODULE-IV

- X. a) Explain the X-ray and IR testing of tyres. (8)

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b) Explain the tests conducted on tread compound. (7)

OR

XI. Explain the following tests (i) High way testing (ii) holography
(iii) Plunger test. (5X3=15)
